1) Hospital Source Systems (Data Producers)

This is the “ground truth” inside the provider network:

- EHR/HIS (encounters, diagnoses, vitals, notes)
- LIS (lab results), RIS/PACS (imaging reports), pharmacy systems
- claims/billing/RCM, scheduling, devices/ICU feeds, registries (if any)

Output: raw clinical/operational data generated during real care.

2) Interoperability & Integration Layer (Connectivity)

This layer connects everything without disrupting operations:

- HL7/FHIR interfaces, APIs, integration engine
- ETL/ELT pipelines, event streaming where needed
- identity resolution (patient matching), provider & facility identifiers

Output: consistent, automated data flows from multiple systems into a unified pipeline.

3) RWD Foundation (Curation, Standardization, and Data Quality)

Here we transform raw hospital data into research-ready RWD:

- mapping to common models (FHIR resources and/or OMOP)
- terminology normalization (ICD, SNOMED, LOINC, RxNorm)
- data quality checks, completeness scoring, lineage & versioning
- de-identification/pseudonymization rules and access tiers

Output: standardized RWD assets with quality, traceability, and controlled access.

4) Research Operations Enablement (CRO Capabilities)

This is how the provider network becomes “trial-ready”:

- eIRB workflows (ethics submissions, approvals, amendments)
- CTMS (site management, budgets, milestones, monitoring)
- EDC (study data capture), ePRO/eCOA (patient-reported outcomes if needed)
- eFeasibility (can we recruit enough eligible patients?)
- eRecruitment (patient identification + outreach workflows under governance)

Output: operational infrastructure to run studies efficiently and compliantly.

5) AI & Clinical Enrichment (NLP + Phenotyping + Multi-modal Integration)

This layer increases the value of the dataset and enables advanced cohorts:

- NLP on clinical notes (extract phenotypes, symptoms, severity, timelines)
- cohort building (inclusion/exclusion logic at scale)
- endpoint detection (events, complications, readmissions)
- genetics + clinical linkage (when permitted and available)
- biomarker discovery support (signals + hypothesis generation)

Output: enriched features, cohorts, and endpoints—higher signal RWD.

6) Evidence Factory (RWE Analytics + VBHC/VBP Measurement)

This layer turns data into evidence and outcomes:

- study templates (protocol → analysis plan → reporting)
- comparative effectiveness, safety, utilization, adherence
- quality measures, outcomes dashboards, risk adjustment & attribution
- VBHC enablement: continuous measurement loops to improve pathways
- VBP enablement: contract metrics (KPIs), baselines, shared-savings logic

Output: evidence packages + performance measurement products for providers, payers, regulators, and industry.

7) Preprint & Knowledge Publishing Platform (Scientific Exchange Layer)

This is your ecosystem's "knowledge distribution engine":

- preprints with versioning, DOIs, ORCID, metadata standards
- links to protocol/IRB reference, cohort definitions, code lists, methods
- reproducibility artifacts (where possible): analysis code, synthetic/aggregate outputs
- governance-aware publishing (no raw patient data; controlled disclosures)

Output: credible, traceable scientific outputs that accelerate collaboration, visibility, and fundraising.

We connect hospital systems, standardize RWD, enable research operations, enrich data with AI, produce RWE outcomes that power VBHC and VBP, then publish evidence through a governed preprint platform to scale collaboration and credibility.